论文首页与 OpenReview 录用通知:

Training-Free Hierarchical Scene Understanding for Gaussian Splatting with Superpoint Graphs

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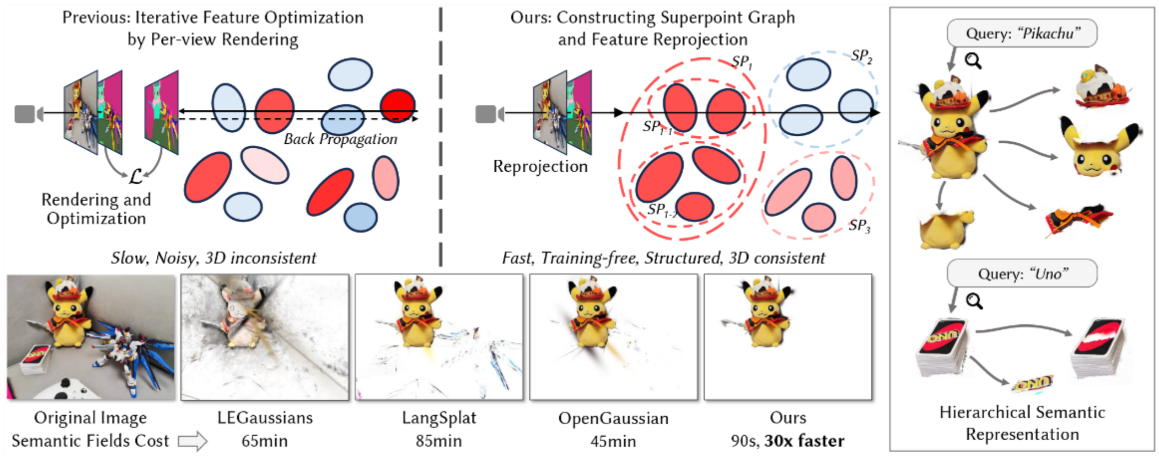

Figure 1: We propose a method for open-vocabulary 3D scene understanding by integrating Gaussian Splatting with a superpoint graph. The top row shows that existing methods iteratively optimize features over Gaussian primitives, resulting in slow processing and inconsistent 3D semantics. In contrast, our approach clusters Gaussians into superpoints (denoted as *SP*) and uses feature reprojection to build a semantic field. The bottom row highlights 3D consistency and speed improvements. The examples on the right demonstrate the capability of our work for open-vocabulary and hierarchical scene understanding.

Abstract

Bridging natural language and 3D geometry is a crucial step toward flexible, language-driven scene understanding. While recent advances in 3D Gaussian Splatting (3DGS) have enabled fast and high-quality scene reconstruction, research has also explored incorporating open-vocabulary understanding into 3DGS. However, most existing methods require iterative optimization over per-view 2D semantic feature maps, which not only results in inefficiencies but also leads to inconsistent 3D semantics across views. To address these limitations, we introduce a training-free framework that constructs a superpoint graph directly from Gaussian primitives. The superpoint graph partitions the scene into spatially compact and semantically coherent regions, forming view-consistent 3D entities and providing a structured foundation for open-vocabulary understanding. Based on the graph structure, we design an efficient reprojection strategy that lifts 2D semantic features onto the superpoints, avoiding costly multi-view iterative training. The resulting representation ensures strong 3D semantic coherence and naturally supports hierarchical understanding, enabling both coarse- and fine-grained open-vocabulary perception within a unified semantic field. Extensive experiments demonstrate that our method achieves state-of-the-art open-vocabulary segmentation performance, with semantic field reconstruction completed over 30× faster. Our code will be available at https://github.com/Atrovast/THGS.

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CCS Concepts

• **Computing methodologies → Scene understanding**.

Keywords

Open-vocabulary, Scene Understanding, Gaussian Splatting, Superpoint

1 Introduction

In recent years, computer vision has made significant progress, particularly in developing systems that perceive and interact with the three-dimensional world. One emerging challenge in this context is 3D open-vocabulary scene understanding, where machines are expected to identify and localize arbitrary regions within a 3D environment based on free-form natural language input. This capability plays an important role in applications such as augmented reality and robotics, enabling users to refer to objects or regions in a scene

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